Notes: Bennett, Sias, and Starks (RFS, 2003)

Greener Pastures and the Impact of Dynamic Institutional Preferences

Contents

1 Big picture

1.1 Question

The paper studies whether institutional investors' portfolio preferences are stable over time, how any preference shifts arise, whether they affect market outcomes, and why institutional investors might migrate toward different stocks. The central question is not just whether institutions prefer large and liquid firms in the cross section, but whether that preference changes dynamically as institutional ownership grows.

1.2 Setting

The empirical setting is the U.S. equity market from March 1983 through December 1997. Institutional ownership comes from quarterly 13F filings, and firm characteristics come from CRSP. The paper separates institutional investors into five groups: independent investment advisers, bank trust departments, mutual funds, insurance companies, and unclassified institutions.

The sample period is important because institutional ownership rises sharply. The paper reports that institutions held about 7% of U.S. equities in 1950, 28% in 1970, and more than 50% by 1999. Thus the period is one in which institutions increasingly become the marginal investors in many stocks.

1.3 Core idea

The authors measure institutional preferences by regressing institutional ownership on firm characteristics each quarter. The characteristics include risk, size, age, dividend yield, price, turnover, and lag return. Because both institutional ownership and firm characteristics change scale over time, the paper standardizes all variables each quarter. A standardized coefficient is interpreted as the expected standard-deviation change in institutional ownership associated with a one standard-deviation change in a stock characteristic.

1.4 Main results

The paper's main findings are:

- Institutional investors still prefer large, liquid stocks, but this preference becomes weaker over time.
- Institutions move toward smaller and riskier securities, especially in the 1990s.
- The shift in aggregate preferences is mostly due to changes in the preferences of each institutional class, not merely changes in the relative size of mutual funds, banks, advisers, and insurers.
- The growth and changing preferences of institutional investors help explain increases in turnover and firm-specific volatility, especially among small stocks.
- Smaller stocks appear to offer "greener pastures": relatively better valuations and more opportunity for informed institutional trading.

1.5 Economic takeaway

The paper reframes institutional ownership as an active, time-varying force in asset markets. Institutions are not simply a group with fixed demand for large liquid firms. As they become larger and as large stocks become crowded, they migrate toward smaller and riskier firms. This migration matters because it changes liquidity, volatility, and the cross section of future returns.

2 Incremental contribution of the paper

2.1 Relative to earlier institutional preference papers

Earlier work such as Del Guercio (1996), Falkenstein (1996), and Gompers and Metrick (2001) studies institutional ownership in the cross section and documents that institutions prefer large, liquid, high-price, seasoned stocks. Bennett, Sias, and Starks extend this literature in four ways.

First, the paper makes institutional preferences explicitly dynamic. The authors do not treat the relation between institutional ownership and firm characteristics as fixed. They estimate quarterly cross-sectional preference coefficients and then study how those coefficients evolve.

Second, the paper decomposes aggregate institutional preferences into two components: the preferences of each investor class and the impact of each class on aggregate ownership. This distinction matters because aggregate institutional ownership can change either because mutual funds become more important or because all classes of institutions change their own stock preferences.

Third, the paper links institutional preference shifts to market outcomes. It studies whether changes in institutional ownership forecast future turnover and firm-specific risk. This moves the paper beyond portfolio choice and toward the consequences of institutionalization for market structure.

Fourth, the paper investigates why preferences shift. It tests two "greener pastures" channels: demand shocks that make large firms expensive, and information advantages that are more valuable in smaller firms.

2.2 Why the contribution is incremental

The paper's novelty is not merely documenting that institutions like large firms. The incremental contribution is to show that this preference weakens over time, that the change comes primarily from within-class preference changes, and that the resulting shift plausibly contributes to broader changes in liquidity, volatility, and return predictability.

3 Data and measurement

3.1 Institutional ownership data

Institutional ownership is measured from CDA Spectrum 13F filings. The sample covers 60 quarters from March 1983 to December 1997. The data classify institutional investors into five types:

1. independent investment advisers;

2. bank trust departments;
3. mutual funds;
4. insurance companies;
5. unclassified institutional investors.

For each stock-quarter, the fraction of shares held by all institutions is the sum of the ownership shares across the five institutional categories. The paper also separately analyzes the ownership share of each category.

3.2 Firm characteristics

The paper studies nine stock characteristics:

- **Risk**: beta, return standard deviation, firm-specific risk.
- **Investment constraints and prudence**: firm size, firm age, dividend yield.
- **Liquidity**: share price and turnover.
- **Momentum control**: lag six-month return.

Beta is estimated with contemporaneous and lagged market returns. Total risk is the standard deviation of past monthly returns. Firm-specific risk is based on daily residual return variation relative to industry returns. Size is the logarithm of equity market capitalization; age is the logarithm of months on CRSP since December 1972; dividend yield, price, and turnover are updated each quarter.

3.3 Important measurement choice: standardization

The paper standardizes both dependent and independent variables each quarter. This is crucial because institutional ownership grows over the sample. Without standardization, raw coefficients could mechanically rise because the dependent variable becomes larger, even if preferences do not change.

The standardized coefficient has a clean interpretation: it measures how many standard deviations institutional ownership changes when a stock characteristic changes by one standard deviation. This makes coefficients comparable across time, characteristics, and institutional categories.

4 Empirical framework: measuring institutional preferences

4.1 Baseline cross-sectional preference regression

In each quarter, the paper estimates a cross-sectional regression of institutional ownership on firm characteristics:

$$y_i = b_0 + b_1 x_{i1} + b_2 x_{i2} + \cdots + b_J x_{iJ} + \varepsilon_i. \tag{1}$$

After standardizing the variables, the regression becomes:

$$\frac{y_i - \bar{y}}{s_y} = \sum_{j=1}^{J} b_j \frac{s_j}{s_y} \left(\frac{x_{ij} - \bar{x}_j}{s_j} \right) + \varepsilon_i^*, \tag{2}$$

where s_y is the cross-sectional standard deviation of institutional ownership and s_j is the cross-sectional standard deviation of characteristic j.

The standardized preference coefficient for characteristic j is:

$$\beta_j = b_j \frac{s_j}{s_y}. \tag{3}$$

4.2 Interpretation of key coefficients

A positive coefficient on size means institutions hold more of larger firms, conditional on all other characteristics. A positive coefficient on standard deviation means institutions tilt toward higher total-risk stocks. A positive coefficient on price or turnover means they prefer more liquid stocks. A negative coefficient on dividend yield means they avoid high-dividend firms after controlling for size, price, risk, and other characteristics.

4.3 Subperiod comparison

The authors split the sample into two periods:

- early period: 1983:Q1 to 1990:Q2;
- late period: 1990:Q3 to 1997:Q4.

They compare distributions of quarterly standardized coefficients using Wilcoxon rank-sum tests. The purpose is to test whether aggregate institutional preferences are stable.

5 Main empirical results: preference shifts

5.1 Aggregate institutional preferences

Over the full sample, institutions prefer large, older, liquid, high-price stocks. They avoid high-dividend-yield stocks after controls. The average coefficient on size is large and positive, and the coefficient on share price is also large and positive. These are the two most important characteristics in magnitude.

The key dynamic finding is that institutions become more willing to hold smaller and riskier stocks in the late period. The size coefficient declines, the standard-deviation coefficient rises sharply, and the dividend-yield coefficient becomes more negative. The overall interpretation is a migration away from very conservative holdings toward riskier, smaller names.

5.2 Not an indiscriminate move into illiquid stocks

The move toward smaller firms is not simply a move into any small stock. The liquidity results are more nuanced. Institutions increase their preference for high-price stocks, but reduce the coefficient on turnover. Since the increase in the price preference dominates, the paper interprets the net effect as a stronger preference for liquidity even as institutions move toward smaller firms.

5.3 Differences across investor classes

Different institutional classes have different preferences. Bank trust departments are more conservative: they have stronger preference for size and age and less appetite for risk. Independent investment advisers and mutual funds are less conservative: they have weaker size preferences and stronger risk preferences.

This heterogeneity matters because the growth of less conservative classes, especially independent advisers and mutual funds, could mechanically affect aggregate preferences. The paper therefore asks whether aggregate changes come from class-size changes or from within-class preference changes.

6 Decomposing aggregate preferences

6.1 Aggregate preference as weighted class preferences

Let $\beta_{\text{total},j}$ be the aggregate standardized preference for characteristic j. Let $\beta_{k,j}$ be the preference of institutional class k and let s_k/s_{total} be that class's preference impact. Then:

$$\beta_{\text{total},j} = \sum_{k=1}^{5} \frac{s_k}{s_{\text{total}}} \beta_{k,j}. \tag{4}$$

This equation says that aggregate institutional preferences can change because $\beta_{k,j}$ changes, because s_k/s_{total} changes, or because the two move together.

6.2 Variance decomposition

The paper decomposes the time-series variation in aggregate preference coefficients. For a given characteristic:

$$\sigma^2(\beta_{\text{total}}) = \sum_{k=1}^{5} \sigma^2\left(\frac{s_k}{s_{\text{total}}}\beta_k\right) + 2\sum_{k=1}^{5}\sum_{m<k} \operatorname{cov}\left(\frac{s_k}{s_{\text{total}}}\beta_k, \frac{s_m}{s_{\text{total}}}\beta_m\right). \tag{5}$$

The paper further partitions each class contribution into:

- the part due to changes in preferences β_k;
- the part due to changes in preference impacts s_k/s_{total};
- the interaction term.

6.3 Main decomposition result

The decomposition is the most important structural result in the paper. On average across characteristics, about 93% of the time-series variation in aggregate preferences comes from changes in the preferences of institutional classes. Only about 4% comes from changes in the relative importance of investor classes, and about 3% from interaction terms.

Economic message. Aggregate institutional migration toward smaller and riskier stocks is not mainly a compositional artifact. It is not just that mutual funds became more important. Rather, each class of institution changed what it wanted to hold.

7 Impact on turnover and firm-specific risk

7.1 Turnover regression

To test whether institutional ownership growth affects liquidity, the paper estimates quarterly regressions of next-quarter turnover on current characteristics and changes in institutional ownership:

$$\text{Turnover}_{i,t+1} = \sum_{j=1}^{9} \beta_{j,t} X_{ij,t} + \beta_{10,t} \Delta\%\text{Total}_{i,t} + \varepsilon_{i,t}. \tag{6}$$

All variables are standardized. The coefficient on $\Delta\%\text{Total}_{i,t}$ tests whether stocks experiencing institutional ownership increases subsequently have higher turnover, after controlling for existing turnover and other characteristics.

7.2 Turnover findings

The coefficient on changes in institutional ownership is positive and statistically significant. This supports the view that institutionalization contributes to growth in trading volume. The authors also show that turnover rises over time for all firms, but the increase is proportionally larger for small firms. The ratio of small-firm turnover to large-firm turnover rises from about 0.80 to 0.93.

7.3 Firm-specific risk regression

The analogous regression for firm-specific risk is:

$$\text{Firm-specific risk}_{i,t+1} = \sum_{j=1}^{9} \beta_{j,t} X_{ij,t} + \beta_{10,t} \Delta\%\text{Total}_{i,t} + \varepsilon_{i,t}. \tag{7}$$

The coefficient on changes in institutional ownership is again positive and statistically significant. This suggests that institutional investors are linked to future firm-specific return variation.

7.4 Why institutionalization can raise firm-specific risk

The paper discusses several mechanisms. Institutional trades are often larger than retail trades, so they can move prices. Institutions may also trade on firm-specific information, which increases idiosyncratic price discovery. Finally, institutional herding can concentrate trading pressure in the same direction, increasing volatility in favored or disfavored stocks.

Economic message. Institutional ownership is not only a passive balance-sheet fact. Increases in institutional ownership are associated with future increases in trading activity and firm-specific volatility. Because institutions move toward smaller stocks, the market impact is especially visible among small-capitalization firms.

8 Why did institutions move toward smaller and riskier stocks?

8.1 The "greener pastures" hypothesis

The paper offers two related explanations. First, institutional demand for large stocks may have raised large-stock valuations, making smaller stocks relatively more attractive. Second, smaller stocks may offer better opportunities for institutions to use informational advantages because they are less efficiently priced and less heavily competed over by other informed traders.

8.2 Large-stock valuation channel

The paper studies book-to-market ratios across small and large firms. Large firms experience a significant decline in book-to-market ratios between the early and late periods, while small firms do not show a comparable decline. The ratio of small-firm to large-firm book-to-market rises significantly. This is consistent with large firms becoming relatively expensive as institutional demand for them grows.

8.3 Institutional demand shocks and informed trading

The paper estimates future-return regressions of the form:

$$\text{Return}_{i,t+1} = \sum_{j=1}^{9} \beta_{j,t} X_{ij,t} + \beta_{10,t} \%\text{Total}_{i,t-1} + \beta_{11,t} \Delta\%\text{Total}_{i,t} + \varepsilon_{i,t}. \tag{8}$$

A positive coefficient on beginning-of-quarter institutional ownership supports the demand-shock view: high institutional ownership predicts future returns. The paper finds such a relation.

However, changes in institutional ownership as a simple share change do not strongly forecast returns. To measure informed trading better, the paper uses a signed herding measure based on Lakonishok, Shleifer, and Vishny (1992).

8.4 Signed herding measure

For stock i in quarter t, define:

$$H_{i,t} = \left| \frac{B_{i,t}}{B_{i,t} + S_{i,t}} - p_t \right| - AF_{i,t}, \tag{9}$$

where $B_{i,t}$ is the number of institutions increasing positions, $S_{i,t}$ is the number decreasing positions, p_t is the average buy ratio in the cross section, and $AF_{i,t}$ is an adjustment factor. The signed herding measure is positive when the stock is buy-herded and negative when it is sell-herded.

The paper finds that signed herding predicts future returns, even after controlling for characteristics and institutional ownership levels. This supports the view that institutions possess information or skill in selecting stocks.

8.5 Small versus large stocks

The return spread between heavily bought and heavily sold stocks is much larger for small stocks than for large stocks. For small firms, the top-buy-herding portfolio outperforms the top-sell-herding portfolio by about 5.09% over the next four quarters. For large firms, the corresponding spread is only about 1.17% and is not statistically significant.

Economic message. Institutions appear to move toward smaller stocks because the expected payoff to research and informed trading is higher there. The market for large stocks may be too crowded and expensive; smaller stocks offer less crowded information rents.

9 Identification, interpretation, and limitations

9.1 What the design identifies

The paper documents conditional relations: how institutional ownership, preferences, turnover, volatility, and returns move together over time and across stocks. The standardized regressions are descriptive but informative because they show whether the characteristics associated with institutional demand shift in systematic ways.

9.2 Why the evidence is persuasive

The main strength is the decomposition. Without it, the observed shift toward smaller and riskier stocks could be explained by the growth of mutual funds. The decomposition shows this is not the dominant mechanism. Most of the shift arises from changes within investor classes.

The paper also triangulates with several outcomes. Preference changes line up with higher turnover, higher firm-specific risk, valuation shifts, and future return predictability. The results therefore form a coherent story rather than a single regression result.

9.3 Limitations

The paper does not provide a randomized source of variation in institutional ownership. Some regressions may capture equilibrium relations rather than causal effects. For example, institutional ownership could forecast turnover because institutions select stocks that are about to become liquid, rather than causing liquidity. Similarly, signed herding could capture correlated information signals rather than trading skill in a strict causal sense.

The sample ends in 1997, before the later rise of ETFs, high-frequency trading, and passive investing. The economic mechanisms remain relevant, but the institutional landscape has changed substantially since the sample period.

10 Extracted figures and tables from the paper

10.1 Table 1: Descriptive statistics; Figure 1: Institutional ownership by institutional investor type over time

Table 1
Descriptive statistics

	Mean	Median	Minimum	Maximum
Panel A: Percentage institutional ownership				
% Total	0.2277	0.2179	0.1544	0.3093
% Ind. inv. advisers	0.1141	0.1174	0.0509	0.1583
% Bank trust depts.	0.0515	0.0507	0.0400	0.0620
% Mutual funds	0.0286	0.0170	0.0115	0.0715
% Insurance	0.0184	0.0175	0.0153	0.0231
% Unclassified	0.0151	0.0151	0.0110	0.0200
% Market capitalization	0.4462	0.4500	0.3430	0.5258
Panel B: Share characteristics				
Beta	1.28	1.30	1.03	1.58
Standard deviation	13.82%	13.87%	12.62%	15.05%
Firm-specific risk	4.34%	4.20%	1.67%	10.03%
Size ($000)	687,345	604,326	351,821	1,519,337
Age (months)	124	121	103	139
Dividend (%/month)	0.16	0.15	0.13	0.25
Price ($)	18.90	18.40	12.70	28.57
Turnover	6.50%	5.90%	3.67%	11.08%
Lag 6-month return	8.35%	8.16%	−25.99%	52.21%
Number of firms	5,425	5,449	4,345	6,779

Each quarter between March 1983 and December 1997 (60 quarters) we estimate the cross-sectional average fraction of shares held by institutional investors (overall and by type) and share characteristics. Beta is estimated as the sum of the coefficients in a regression of the firm's monthly return on the contemporaneous and lag one-month CRSP NYAM value-weighted index over the previous 24 to 60 months (depending on availability). Standard deviation is estimated from monthly returns over the previous 24 to 60 months (depending on availability). Firm-specific risk is estimated as the squared difference between daily firm returns and associated industry returns summed each month and averaged over the three months in the current quarter. Firm age is given as the number of months since December 1972 for which the firm has CRSP return data. Dividend yield is measured as the average monthly dividend yield over the previous 12 months. Share price is given as the quarter-end share price. Turnover is defined as the ratio of monthly volume to number of shares outstanding averaged over the three months in the current quarter. Lag return is the cumulative return for the firm over the previous six months. The time-series mean, median, minimum, and maximum of the 60 quarterly cross-sectional averages are reported above.

(a) Table 1: Descriptive statistics

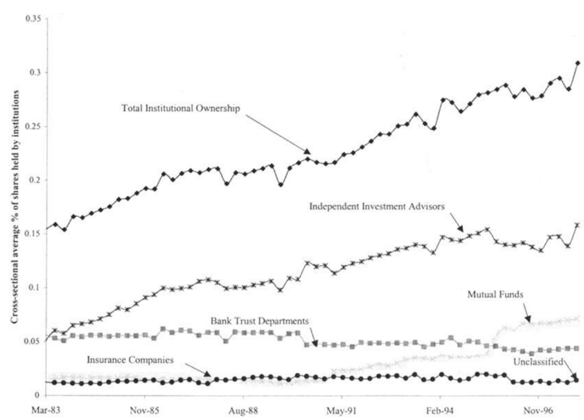

Figure 1.

(b) Figure 1: Institutional ownership by institutional investor type over time

Read.

- Table 1 shows that total institutional ownership averages 22.77% of shares and rises to a maximum of 30.93% by the end of the sample.
- The largest institutional categories are independent investment advisers and bank trust departments.
- Figure 1 shows total institutional ownership rising steadily, with growth especially driven by independent investment advisers and, later, mutual funds.

Economic message. The market becomes increasingly institutionalized over the sample. The rising institutional share creates the economic importance of asking whether institutions' portfolio demands change over time.

10.2 Table 2: Average cross-sectional correlations between institutional ownership and share characteristics; Table 3: Tests for changes in aggregate preferences

Table 2
Average cross-sectional correlations between institutional ownership and share characteristics

	Beta	Standard deviation	Firm-specific risk	Size	Age	Dividend yield	Price	Turnover	Lag return
Panel A: Correlation between institutional ownership and share characteristics									
% Total	0.0381	−0.2652	−0.2472	0.6533	0.2884	0.0316	0.5973	0.2476	0.0508
% Ind. inv. advisers	0.0852	−0.1571	−0.2091	0.4847	0.1973	−0.0335	0.4754	0.2815	0.0538
% Bank trust depts.	−0.0560	−0.3087	−0.1892	0.5269	0.3001	0.1078	0.4859	0.0626	0.0309
% Mutual funds	0.0408	−0.1353	−0.1516	0.4131	0.1350	0.0060	0.3757	0.2010	0.0431
% Insurance	0.0048	−0.1424	−0.1157	0.3672	0.1658	0.0364	0.2897	0.0985	0.0112
% Unclassified	0.0114	−0.1488	−0.1174	0.3869	0.1694	0.0293	0.3007	0.0920	−0.0016
Panel B: Share characteristics									
Beta	1.0000								
Standard deviation	0.4593	1.0000							
Firm-specific risk	0.1218	0.4070	1.0000						
Size	−0.0469	−0.5006	−0.3794	1.0000					
Age	−0.1216	−0.2753	−0.1580	0.3254	1.0000				
Dividend yield	−0.2384	−0.4870	−0.1641	0.2340	0.1268	1.0000			
Price	−0.1457	−0.6346	−0.4731	0.8048	0.3230	0.2517	1.0000		
Turnover	0.1974	0.2114	0.0444	0.1427	−0.0978	−0.1168	0.1167	1.0000	
Lag 6-month return	0.0138	−0.0112	−0.0869	0.1445	0.0431	0.0306	0.2229	0.1393	1.0000

Each quarter between March 1983 and December 1997 we estimate the cross-sectional correlation between share characteristics and the fraction of shares held by institutional investors (in aggregate and by investor classification). This table presents the time-series average of the 60 quarterly cross-sectional correlations. Panel A presents the mean correlation between share characteristics and the institutional ownership measures (overall and by type). Panel B presents the mean correlations between the share characteristics. Beta is estimated as the sum of the coefficients in a regression of the firm's monthly return on the contemporaneous and lag one-month CRSP NYAM value-weighted index over the previous 24 to 60 months (depending on availability). Standard deviation is estimated as the natural logarithm of monthly return standard deviation over the previous 24 to 60 months (depending on availability). Firm-specific risk is estimated as the natural logarithm of one plus the average monthly estimate of firm-specific risk in the current quarter. The monthly estimate is generated as the squared difference between daily firm returns and associated industry returns summed each month. Firm size is measured as the natural logarithm of equity capitalization. Firm age is given as the natural logarithm of the number of months since December 1972 for which the firm has CRSP return data. Dividend yield is measured as the natural logarithm of one plus the average monthly dividend yield over the previous 12 months. Share price is given as the natural logarithm of one plus the quarter-end share price. Turnover is defined as the natural logarithm of one plus the monthly turnover averaged over the three months in the current quarter. Monthly turnover is given as the ratio of monthly volume to number of shares outstanding. Lag return is the cumulative return for the firm over the previous six months.

(a) Table 2: Average cross-sectional correlations between institutional ownership and share characteristics

Table 3
Tests for changes in aggregate preferences

Period	Beta	Standard deviation	Firm specific risk	Size	Age	Dividend yield	Price	Turnover	Lag return
Panel A: Entire sample period									
831–974 ($n=60$)	0.0372 (0.73/0.89) (0.27/0.69)	0.0681 (0.78/0.75) (0.22/0.31)	0.0100 (0.72/0.44) (0.28/0.12)	0.4377 (1.00/1.00) (0.00/.)	0.1012 (1.00/0.98) (0.00/.)	−0.0920 (0.00/.) (1.00/0.95)	0.2912 (1.00/1.00) (0.00/.)	0.1378 (1.00/1.00) (0.00/.)	−0.0987 (0.00/.) (1.00/1.00)
Panel B: Subperiod analysis									
831–902 ($n=30$)	0.0415 (0.77/0.87) (0.23/0.43)	0.0037 (0.57/0.29) (0.43/0.31)	0.0070 (0.60/0.33) (0.40/0.08)	0.4920 (1.00/1.00) (0.00/.)	0.0967 (1.00/0.97) (0.00/.)	−0.0716 (0.00/.) (1.00/0.90)	0.2081 (1.00/1.00) (0.00/.)	0.1464 (1.00/1.00) (0.00/.)	−0.1018 (0.00/.) (1.00/1.00)
903–974 ($n=30$)	0.0328 (0.70/0.90) (0.30/0.89)	0.1324 (1.00/1.00) (0.00/.)	0.0129 (0.83/0.52) (0.17/0.20)	0.3835 (1.00/1.00) (0.00/.)	0.1058 (1.00/1.00) (0.00/.)	−0.1124 (0.00/.) (1.00/1.00)	0.3743 (1.00/1.00) (0.00/.)	0.1292 (1.00/1.00) (0.00/.)	−0.0957 (0.00/.) (1.00/1.00)
z-statistic	−0.72	6.65**	1.77	−5.40**	0.95	−4.50**	6.63**	−2.62**	0.75

Each quarter between March 1983 and December 1997 we estimate a cross-sectional regression of the fraction of shares held by institutional investors on nine share characteristics. The number of securities in each cross-sectional regression ranges from 4345 to 6779. Both the level of institutional ownership and share characteristics are standardized each quarter, that is, rescaled to have a mean of zero and a standard deviation of one. The first number in each cell reports the time-series average coefficient from the quarterly regressions. The second row in each cell reports the fraction of the regressions with a positive coefficient associated with the characteristic followed by the fraction of positive coefficients that are statistically significant [at the 5% level or better based on White's (1980) heteroscedasticity-consistent covariance estimator]. The third row in each cell reports the fraction of the regressions with a negative coefficient associated with that characteristic followed by the fraction of negative coefficients that are statistically significant. Results are reported for the entire sample period (panel A) and for two subperiods (panel B): the first quarter of 1983 through the second quarter of 1990 and the third quarter of 1990 through the fourth quarter of 1997. The last row presents the results of a Wilcoxon ranked-sum test of the null hypothesis that the coefficient estimates in the first period equal the coefficient estimates in the second period. The average R^2 for the 60 cross-sectional regressions is 51.05%. All variables are as defined in Table 2. ** indicates statistical significance at the 1% level; * indicates statistical significance at the 5% level.

(b) Table 3: Tests for changes in aggregate preferences

Read.

- Table 2 shows strong positive pairwise correlations of institutional ownership with size and price, and negative correlations with risk measures such as standard deviation and firm-specific risk.
- Table 3 moves to multivariate standardized regressions and shows that institutions prefer size, age, price, and turnover, while avoiding dividend yield and high lag returns.
- The subperiod comparison in Table 3 shows a large increase in preference for standard deviation and a decline in preference for size.

Economic message. Pairwise correlations show the traditional institutional preference for large, liquid stocks. The standardized multivariate regressions reveal the dynamic shift: by the 1990s, institutions become more willing to hold riskier and smaller securities.

10.3 Table 4: Regressions of standardized ownership by institutional classification on standardized share characteristics; Table 5: Tests for changes in preferences and preference impact over time by investor class

Read.

- Table 4 shows that different institutional classes have heterogeneous preferences. Bank trust departments are relatively conservative; advisers and mutual funds are less conservative.
- Table 5 shows that both preference impacts and preferences change over time.

Table 4
Regressions of standardized ownership by institutional classification on standardized share characteristics

	Independent investment advisors	Bank trust departments	Mutual funds	Insurance companies	Unclassified	Average p-value
Beta	0.0615	−0.0029	0.0196	0.0023	0.0041	0.1203
	(0.82/0.84)	(0.38/0.49)	(0.82/0.59)	(0.57/0.53)	(0.55/0.55)	(0.65)
	(0.18/0.45)	(0.62/0.38)	(0.18/0.82)	(0.43/0.46)	(0.45/0.19)	
Standard deviation	0.0800	−0.0103	0.0749	0.0367	0.0349	0.1101
	(0.95/0.65)	(0.53/0.34)	(0.88/0.64)	(0.85/0.55)	(0.85/0.67)	(0.68)
	(0.05/0.00)	(0.47/0.64)	(0.12/0.00)	(0.15/0.11)	(0.15/0.11)	
Firm-specific risk	−0.0269	0.0505	0.0066	0.0160	0.0234	0.0281
	(0.07/0.25)	(1.00/0.97)	(0.70/0.12)	(0.82/0.57)	(1.00/0.65)	(0.82)
	(0.93/0.63)	(0.00/.)	(0.30/0.00)	(0.18/0.18)	(0.00/.)	
Size	0.2356	0.3546	0.2842	0.3569	0.3925	0.0019
	(1.00/1.00)	(1.00/1.00)	(1.00/1.00)	(1.00/1.00)	(1.00/1.00)	(0.98)
	(0.00/.)	(0.00/.)	(0.00/.)	(0.00/.)	(0.00/.)	
Age	0.0675	0.1282	0.0246	0.0630	0.0600	0.0562
	(0.98/0.92)	(1.00/1.00)	(0.83/0.72)	(1.00/0.90)	(1.00/0.90)	(0.85)
	(0.02/0.00)	(0.00/.)	(0.17/1.00)	(0.00/.)	(0.00/.)	
Dividend yield	−0.1069	−0.0357	−0.0497	−0.0327	−0.0460	0.0125
	(0.00/.)	(0.05/0.00)	(0.05/0.00)	(0.05/0.00)	(0.00/.)	(0.97)
	(1.00/1.00)	(0.95/0.68)	(0.95/0.68)	(0.95/0.74)	(1.00/0.92)	
Price	0.3305	0.2010	0.2020	0.0274	0.0224	0.0001
	(1.00/1.00)	(1.00/1.00)	(1.00/0.93)	(0.62/0.49)	(0.53/0.56)	(1.00)
	(0.00/.)	(0.00/.)	(0.00/.)	(0.38/0.00)	(0.47/0.07)	
Turnover	0.1832	0.0054	0.1189	0.0456	0.0339	0.0001
	(1.00/1.00)	(0.50/0.33)	(1.00/1.00)	(0.95/0.61)	(0.90/0.67)	(1.00)
	(0.00/.)	(0.50/0.20)	(0.00/.)	(0.05/0.00)	(0.10/0.33)	
Lag return	−0.0821	−0.0677	−0.0569	−0.0526	−0.0693	0.1178
	(0.00/.)	(0.00/.)	(0.02/0.00)	(0.00/.)	(0.00/.)	(0.65)
	(1.00/0.95)	(1.00/1.00)	(0.98/0.92)	(1.00/0.95)	(1.00/0.92)	
Average R^2	34.77%	31.89%	22.15%	15.19%	16.98%	

Each quarter between March 1983 and December 1997 we estimate cross-sectional regressions of fractional institutional ownership (for each class of institutional investor) on nine share characteristics. The number of securities in each cross-sectional regression ranges from 4345 to 6779. Both the institutional ownership measures and share characteristics are standardized each quarter, that is, rescaled to have a mean of zero and a standard deviation of one. The first number in each cell reports the time-series average coefficient from the 60 quarterly regressions. The second row in each cell reports the fraction of the 60 regressions with a positive coefficient associated with the characteristic followed by the fraction of positive coefficients that are statistically significant [at the 5% level or better based on White's (1980) heteroscedasticity-consistent covariance estimator]. The third row in each cell reports the fraction of the 60 regressions with a negative coefficient associated with that characteristic followed by the fraction of negative coefficients that are statistically significant. The average R^2 from the 60 regressions are reported in the last row. Each quarter, we also compute an F-statistic associated with a likelihood ratio test of the hypothesis that the coefficients associated with that characteristic are equal for independent investment advisors, bank trust departments, mutual funds, insurance companies, and unclassified

(a) Table 4: Regressions of standardized ownership by institutional classification on standardized share characteristics

Table 5
Tests for changes in preferences and preference impact over time by investor class

	Preference impact	Beta	Standard deviation	Firm-specific risk	Size	Age	Dividend yield	Price	Turnover	Lag return
Panel A: Independent investment advisors										
831–902 ($n=30$)	0.5118	0.0720	0.0290	−0.0206	0.2852	0.0615	−0.0871	0.2797	0.2049	−0.0923
903–974 ($n=30$)	0.5524	0.0509	0.1311	−0.0331	0.1860	0.0735	−0.1268	0.3812	0.1616	−0.0719
z-statistic	3.41**	−1.43	6.02**	−1.29	−5.52**	1.43	−3.98**	6.51**	−3.91**	2.14*
Panel B: Bank trust departments										
831–902 ($n=30$)	0.3805	−0.0012	−0.0435	0.0369	0.4127	0.1006	−0.0200	0.1448	0.0167	−0.0653
903–974 ($n=30$)	0.2810	−0.0046	0.0229	0.0642	0.2965	0.1557	−0.0513	0.2573	−0.0058	−0.0701
z-statistic	−6.56**	−0.41	5.06**	5.58**	−6.19**	4.24**	−5.30**	5.74**	−3.79**	−1.10
Panel C: Mutual funds										
831–902 ($n=30$)	0.1513	0.0238	0.0173	0.0018	0.2624	0.0391	−0.0437	0.1318	0.1141	−0.0557
903–974 ($n=30$)	0.2529	0.0154	0.1326	0.0115	0.3058	0.0101	−0.0557	0.2723	0.1238	−0.0580
z-statistic	5.97**	−0.26	6.44**	2.96**	1.69	−1.66	−0.85	6.54**	1.60	−0.21
Panel D: Insurance companies										
831–902 ($n=30$)	0.1956	0.0023	0.0116	0.0018	0.3531	0.0667	−0.0268	−0.0105	0.0696	−0.0546
903–974 ($n=30$)	0.1708	0.0023	0.0618	0.0303	0.3608	0.0594	−0.0385	0.0653	0.0216	−0.0507
z-statistic	−4.77**	0.07	5.69**	5.82**	1.13	−1.25	−2.79**	5.70**	−4.95**	0.67
Panel E: Unclassified institutional investors										
831–902 ($n=30$)	0.1918	−0.0020	0.0197	0.0159	0.4234	0.0472	−0.0433	−0.0236	0.0258	−0.0589
903–974 ($n=30$)	0.1473	0.0103	0.0502	0.0308	0.3615	0.0728	−0.0488	0.0685	0.0421	−0.0796
z-statistic	−5.57**	2.68**	3.51**	4.22**	−3.64**	3.36**	−1.91	5.43**	1.77	−2.91**

Each quarter between March 1983 and December 1997 we compute the "preference impact" of each institutional investor class as the ratio of the cross-sectional standard deviation of holdings by that investor class to the cross-sectional standard deviation of aggregate institutional holdings. We then report the mean preference impact of each institutional investor class for two subperiods: The first quarter of 1983 through the second quarter of 1990 and the third quarter of 1990 through the fourth quarter of 1997. These averages are reported in the first column. We also present the results of a Wilcoxon ranked-sum test of the null hypothesis that the preference impacts in the first period equal the preference impacts in the second period. In addition, we estimate quarterly cross-sectional regressions of institutional ownership for each institutional investor class on nine share characteristics. Both the institutional ownership measures and share characteristics are standardized each quarter, that is, rescaled to have a mean of zero and a standard deviation of one. We report the mean coefficient for each institutional investor class for the two subperiods as well as the results of a Wilcoxon ranked-sum test of the null hypothesis that the coefficient estimates in the first period equal the coefficient estimates in the second period. All variables are as defined in Table 2. ** indicates statistical significance at the 1% level; * indicates statistical significance at the 5% level.

(b) Table 5: Tests for changes in preferences and preference impact over time by investor class

- Most institutional classes increase their preference for return standard deviation and share price, while several reduce their preference for firm size.

Economic message. The aggregate shift is not only a mutual-fund-growth story. Even within institutional categories, investors change what they want to hold.

10.4 Table 6: Explaining time-series variation in aggregate preferences; Table 6 continued: Aggregate component of the variance decomposition

Table 6
Explaining time-series variation in aggregate preferences

	Beta	Standard deviation	Firm-specific risk	Size	Age	Dividend yield	Price	Turnover	Lag return	Average
Panel A: Independent investment advisors										
Total	0.6475	0.4390	0.4935	0.2845	0.4162	0.6021	0.4031	0.8397	0.6048	0.5256
Preferences	0.5932	0.3766	0.4328	0.3623	0.3499	0.6122	0.2834	0.8706	0.6003	0.4979
Impact	0.0033	0.0019	0.0162	−0.0220	−0.0053	0.0100	0.0103	0.0748	−0.0019	0.0097
Interaction	0.0511	0.0606	0.0444	−0.0557	0.0716	−0.0202	0.1093	−0.1056	0.0063	0.0180
Panel B: Bank trust departments										
Total	0.1298	0.1791	0.2187	0.5664	0.2753	0.1302	0.1526	0.1555	0.1123	0.2133
Preferences	0.1255	0.1380	0.2228	0.2729	0.2907	0.1546	0.2261	0.1765	0.1017	0.1899
Impact	0.0002	0.0006	0.0510	0.0644	0.0324	−0.0032	−0.0096	−0.0042	0.0088	0.0156
Interaction	0.0040	0.0405	−0.0552	0.2292	−0.0478	−0.0213	−0.0639	−0.0168	0.0018	0.0078
Panel C: Mutual funds										
Total	0.0955	0.2784	0.0692	−0.0378	0.1481	0.1685	0.2675	−0.1237	0.1364	0.1113
Preferences	0.0796	0.1607	0.1011	0.0668	0.0855	0.1294	0.1579	0.0594	0.1402	0.1089
Impact	0.0008	0.0042	−0.0047	−0.0423	−0.0068	0.0004	0.0071	0.0757	−0.0093	0.0028
Interaction	0.0150	0.1135	−0.0272	−0.0623	0.0695	0.0387	0.1026	−0.2588	0.0055	−0.0004
Panel D: Insurance companies										
Total	0.0865	0.0691	0.1198	0.0652	0.0717	0.0624	0.0768	0.0377	0.0779	0.0741
Preferences	0.0830	0.0664	0.1480	0.0459	0.0529	0.0732	0.0734	0.1121	0.0811	0.0818
Impact	0.0000	−0.0004	0.0031	0.0172	0.0043	−0.0002	−0.0003	−0.0070	0.0024	0.0021
Interaction	0.0035	0.0031	−0.0314	0.0021	0.0145	−0.0106	0.0037	−0.0674	−0.0056	−0.0098
Panel E: Unclassified										
Total	0.0408	0.0344	0.0989	0.1216	0.0887	0.0368	0.0999	0.0907	0.0686	0.0756
Preferences	0.0369	0.0346	0.1030	0.0303	0.0796	0.0453	0.0880	0.0229	0.0630	0.0560
Impact	−0.0001	−0.0010	0.0119	0.0410	0.0082	−0.0015	−0.0005	−0.0128	0.0059	0.0057
Interaction	0.0040	0.0008	−0.0159	0.0504	0.0009	−0.0070	0.0125	0.0806	−0.0004	0.0140

(a) Table 6: Explaining time-series variation in aggregate preferences

Table 6
Continued

	Beta	Standard deviation	Firm-specific risk	Size	Age	Dividend yield	Price	Turnover	Lag return	Average
Panel F: Aggregate										
Total	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Preferences	0.9182	0.7763	1.0077	0.7781	0.8586	1.0147	0.8289	1.2415	0.9864	0.9345
Impact	0.0041	0.0052	0.0776	0.0583	0.0328	0.0057	0.0069	0.1264	0.0060	0.0359
Interaction	0.0777	0.2185	−0.0852	0.1636	0.1087	−0.0204	0.1641	−0.3679	0.0076	0.0296

Each quarter between March 1983 and December 1997 we compute the "preference impact" for each institutional investor class as the ratio of the cross-sectional standard deviation of ownership by that institutional investor class to the cross-sectional standard deviation of aggregate institutional ownership. We also compute the "preferences" as the standardized regression coefficients from quarterly cross-sectional regressions of fractional institutional ownership (overall and for each class of institutional investor) on nine share characteristics. Because aggregate preferences can be written as a linear function of the product of each investor classes' preference impacts and preferences, we can directly estimate the fraction of time-series variation in aggregate preferences attributed to each institutional investor class. These estimates are reported in the first row of panels A through E. In addition, by assuming preference impacts (preferences) are constant, we can estimate the fraction of time-series variance in aggregate preferences attributed to changes in preferences (preference impacts). These estimates are reported in the second and third row of each panel. The remaining time-series variation in aggregate preferences is attributed to the interaction between preferences and preference impacts and is reported in the last row of each panel. All variables are as defined in Table 2.

(b) Table 6 continued: Aggregate component of the variance decomposition

Read.

- Table 6 decomposes the time-series variation in aggregate preference coefficients.
- The aggregate panel shows that changes in preferences account for about 93.45% of average variation.
- Changes in preference impacts account for only about 3.59%, and interaction effects account for about 2.96%.

Economic message. The dominant force is within-class preference change. Institutions of many types became more willing to hold smaller and riskier stocks; aggregate change is not primarily mechanical composition change.

10.5 Table 7: Institutional ownership, turnover, and firm-specific risk; Table 8: Institutional ownership, valuation levels, and returns

Table 7
Institutional ownership, turnover, and firm-specific risk

Panel A: Regression results

Independent variables	Turnover$_{t+1}$	Firm-specific risk$_{t+1}$
Beta	0.0139	−0.0060
	(4.64)**	(−1.42)
Standard deviation	0.0763	0.0389
	(19.07)**	(6.36)**
Firm-specific risk	−0.0244	0.6118
	(−6.61)**	(40.11)**
Size	0.0461	0.0004
	(10.90)**	(0.13)
Age	−0.0309	−0.0068
	(−12.66)**	(−3.18)**
Dividend yield	−0.0124	−0.0077
	(−6.40)**	(−4.07)**
Price	0.0316	−0.1270
	(6.64)**	(−17.47)**
Turnover	0.7444	−0.0111
	(150.05)**	(−3.43)**
Lag return	−0.0107	−0.0983
	(−1.77)	(−27.21)**
Change in % total	0.0291	0.0102
	(10.47)**	(5.27)**

	Early (3/83–6/90)	Late (9/90–12/97)	z-statistic
Panel B: Turnover at small and large firms (in percent, $n=60$ quarters)			
All firms	5.1086	7.5140	5.42**
Small firms	4.7888	7.1251	5.11**
Large firms	6.0046	7.6399	4.07**
Small/large	0.7987	0.9318	4.47**
Panel C: Firm-specific risk at small and large firms (in percent, $n=60$ quarters)			
All firms	2.9099	4.8550	6.11**
Small firm	3.5603	6.3367	6.22**
Large firms	0.6456	0.7922	4.59**
Small/large	5.9723	8.2656	3.81**

Each quarter between June 1983 and December 1997 we estimate cross-sectional regressions of turnover in quarter $t+1$ on nine security characteristics in quarter t and the change in the fraction of shares held by institutional investors over quarter t. Both the independent and dependent variables are standardized each quarter, that is, rescaled to zero mean, unit variance. The first column in panel A reports the time-series average coefficient from these 59 cross-sectional regressions and associated t-statistics (computed from time-series standard errors). The second column in panel A reports the results of a similar analysis of firm-specific risk. We also compute the cross-sectional average turnover and firm-specific risk for all firms, small firms (bottom third of NYSE capitalization) and large firms (top third of NYSE capitalization) each quarter. In addition, we compute, each quarter, the ratio of the cross-sectional average small firm turnover (firm-specific risk) to the cross-sectional average large firm turnover (firm-specific risk). Panels B and C reports the time-series average of the quarterly cross-sectional averages over the early period (March 1983–June 1990) and the late period (September 1990–December 1997) for each group for turnover and firm-specific risk, respectively. The last column in panels B and C report the results of a Wilcoxon ranked-sum test of the null hypothesis that values in the early period do not differ from the values in the late period. All variables are as defined in Table 2. ** indicates statistical significance at the 1% level; * indicates statistical significance at the 5% level.

(a) Table 7: Institutional ownership, turnover, and firm-specific risk

Table 8
Institutional ownership, valuation levels, and returns

Panel A: Book : market ratios at small and large firms ($n = 60$ quarters)

	Early (3/83–6/90)	Late (9/90–12/97)	z-statistic
All firms	0.5693	0.5602	−1.16
Small firms	0.5562	0.6094	0.73
Large firms	0.5223	0.4048	−5.82**
Small/large	1.1322	1.5109	6.63**

Panel B: Regression results

Independent variables	**Return$_{t+1}$**	**Return$_{t+1}$**
	0.0061	0.0063
Beta	(0.69)	(0.71)
	−0.0327	−0.0331
Standard deviation	(−2.51)*	(−2.54)*
	−0.0361	−0.0350
Firm-specific risk	(−3.34)**	(−3.26)**
	−0.0233	−0.0226
Size	(−2.47)*	(−2.38)*
	0.0129	0.0134
Age	(2.25)*	(2.33)*
	0.0074	0.0077
Dividend yield	(1.05)	(1.09)
	0.0184	0.0174
Price	(1.46)	(1.39)
	−0.0332	−0.0333
Turnover	(−3.72)**	(−3.74)**
	0.0512	0.0501
Lag return	(5.26)**	(5.19)**
	0.0191	0.0201
Beginning-of-quarter % total	(3.05)**	(3.24)**
	0.0017	
Change in % total	(0.50)	
		0.0116
Signed herding measure		(4.33)**

Each quarter, we compute the cross-sectional average book : market ratio for all firms, small firms (bottom third of NYSE capitalization), and large firms (top third of NYSE capitalization). We also compute, each quarter, the ratio of the cross-sectional average small firm book : market ratio to the cross-sectional average large firm book : market ratio. To control for outliers we define the book : market ratio as the natural logarithm of one plus the Compustat book value to CRSP market value. Panel A reports the time-series average of the quarterly cross-sectional averages over the early (March 1983–June 1990) and late periods (September 1990–December 1997) for each group. The last column in panel A reports the results of a Wilcoxon ranked-sum test of the null hypothesis that values in the early period do not differ from the values in the late period. Each quarter between June 1983 and December 1997 we estimate a cross-sectional regression of return in quarter $t+1$ on nine security characteristics in quarter t, the level of institutional ownership at the beginning of quarter t and the change in the fraction of shares held by institutional investors over quarter t. Both the independent and dependent variables are standardized each quarter, that is, rescaled to zero mean, unit variance. The first column in panel B reports the time-series average coefficients from these 59 cross-sectional regressions and associated t-statistics (computed from time-series standard errors). The second column in panel B reports the results for a similar analysis, but replaces the change in the fraction of shares held by institutional investors with the signed herding measure [the Lakonishok, Shleifer, and Vishny (1992) herding measure signed positive if the ratio of the number of institutions buying the security to the number trading the security that quarter is greater than the cross-sectional average ratio that quarter and negative if the ratio of the number of institutions buying the security to the number trading the security that quarter is less than the cross-sectional average ratio that quarter]. Firms must have at least five institutional traders to be included in the analysis. All variables are as defined in Table 2. **indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

(b) Table 8: Institutional ownership, valuation levels, and returns

Read.

- Table 7 shows that changes in institutional ownership positively forecast next-quarter turnover and firm-specific risk.
- The ratio of small-firm turnover to large-firm turnover rises from 0.7987 to 0.9318, while the ratio of small-firm to large-firm firm-specific risk rises from 5.9723 to 8.2656.
- Table 8 shows that large-firm book-to-market ratios fall significantly, and beginning institutional ownership predicts future returns.

Economic message. Institutionalization is linked to market outcomes. The migration toward smaller stocks coincides with relatively larger increases in small-stock liquidity and idiosyncratic volatility.

10.6 Table 9: Average abnormal returns for top buy-herding portfolio less top sell-herding portfolio

Table 9
Average abnormal returns for top buy-herding portfolio less top sell-herding portfolio

	Quarter −2	Quarter −1	Portfolio formation quarter	Quarter +1	Quarter +2	Quarter +3	Quarter +4	Quarters +1 to +4
	Average return from top-buy herding portfolio less average return on top sell-herding portfolio							
Small firms	8.83	11.80	13.55	2.16	1.99	0.83	−0.52	5.09
	(16.52)**	(13.33)**	(16.29)**	(2.71)**	(2.31)*	(1.00)	(−0.52)	(6.46)**
Large firms	2.99	2.53	7.17	0.80	0.12	−0.02	−0.04	1.17
	(6.88)**	(5.17)**	(11.39)**	(1.44)	(0.21)	(−0.04)	(−0.08)	(0.99)

Each quarter between March 1983 and December 1997 we compute the Lakonishok, Shleifer, and Vishny (1992) "herding measure" for each security with at least five institutional traders. Each security is then classified as either a buy herding security (the ratio of the number of institutions buying the security to the number trading the security that quarter is greater than the cross-sectional average ratio that quarter) or a sell herding security (the ratio of the number of institutions buying the security to the number trading the security that quarter is less than the cross-sectional average ratio that quarter). We then compute the average capitalization-adjusted return difference between those securities most heavily purchased by institutional investors (top buy-herding quintile) and those securities most heavily sold by institutional investors (top sell-herding quintile) for small stocks (bottom third of NYSE capitalization) and large stocks (top third of NYSE capitalization). The time-series average of these differences (and associated t-statistics computed from time-series standard errors) in the two quarters prior to the herding quarter, the herding quarter, and the four quarters following the herding quarter are reported above. Capitalization-adjusted returns are calculated as the difference between the firm's return and the average return for firms within the same capitalization group. **indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 9: Average abnormal returns for top buy-herding portfolio less top sell-herding portfolio

Read.

- Table 9 compares the most heavily institutionally bought stocks with the most heavily institutionally sold stocks.
- For small firms, the buy-herding minus sell-herding portfolio earns 5.09% over the next four quarters and is statistically significant.
- For large firms, the corresponding return spread is only 1.17% and is not statistically significant.

Economic message. The return evidence supports the greener-pastures interpretation. Institutional trading appears more informative in small stocks, where institutions may have more scope to exploit information advantages.

11 Short summary

- Institutions historically preferred large, liquid, seasoned stocks.
- From 1983 to 1997, institutions became more willing to hold smaller and riskier stocks.
- The shift is driven mainly by changes in the preferences of each institutional class, not changes in the relative importance of different institutional classes.
- Institutional ownership changes predict future turnover and firm-specific risk.
- Small stocks show especially large increases in turnover and firm-specific volatility.
- The evidence suggests institutions moved to smaller stocks because large stocks became crowded and expensive, while smaller stocks offered stronger opportunities for informed trading.

12 Conclusion

Bennett, Sias, and Starks show that institutional preferences are dynamic. Institutions remain attracted to large and liquid stocks, but the intensity of that preference weakens over time. The institutional sector moves toward smaller and riskier stocks as ownership grows.

The paper's central decomposition is especially important: about 93% of the change in aggregate preferences is due to changing preferences within institutional classes. This means the migration is broad-based across professional investors rather than merely an artifact of mutual-fund growth.

The market consequences are also central. Institutionalization helps explain increases in liquidity and firm-specific risk, and the effects are strongest for smaller firms. Finally, return and valuation evidence supports the idea that institutions moved toward smaller firms because those firms offered greener pastures: more attractive relative valuations and greater scope for informed trading.

The paper therefore links three facts into one economic narrative: institutional ownership grew, institutional preferences changed, and markets changed with them.